Issue Rankings by Severity

Part Identification Repository API | Ranked view of the thematic synthesis

Total issues	Critical	High	Low
12	4	7	1

Rank 1: Critical

Highest-severity issues in the thematic synthesis.

4 issues

N+1 query degradation	Technical stability and backend performance	Critical
Manual looping exhaustion	Engineer experience and workflow efficiency	Critical
Payload bloat and token waste	Payload efficiency and AI cost	Critical
Shadow workarounds	Security and data governance	Critical

Rank 2: High

Significant issues that compound the critical ones.

7 issues

Thread contention and locking	Technical stability and backend performance	High
Memory growth on large payloads	Technical stability and backend performance	High
Stale or out-of-sync cache data	Data integrity and consistency	High
Rate limiting and throttling	Engineer experience and workflow efficiency	High
Brittle error recovery	Engineer experience and workflow efficiency	High
No dynamic compression	Payload efficiency and AI cost	High
Loss of risk-assessment visibility	Security and data governance	High

Rank 3: Low

Aggregated low-priority items.

1 issue

Wrapper maintenance, field ordering, per-item error reporting	Minor technical debt	Low

All research data is simulated. Severity scale: Critical, High, Medium, Low. Counts reflect the 12 line items in the thematic synthesis. Ranking is by severity tier only. Issues within a tier follow the order of the thematic synthesis and are not ranked against each other.